

Resistance test of Post Quantum Schemes against side-channel attacks

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I Introduction

The invention of quantum computers will be extremely dangerous when all users' passwords that protect personal information can be hacked in several days, hours, or even seconds [1]. Then, this information about bank accounts, private photos, chats, and any other valuable data will be available to the attacker. Improving encryption algorithms or replacing them with new schemes [1]–[4] will prevent this issue. Most proposed encryption algorithms are not tested in the natural environment regarding efficiency, performance, security level, and resistance and do not have united standards. Therefore, vulnerabilities in the encryption schemes would not be detected by the theoretical and mathematical analysis [5]–[14]. This paper aims to experimentally examine the resistance of existing post-quantum algorithms to quantum attacks, decide which scheme will protect personal data in the most reliable way, and detect existing vulnerabilities in given schemes. The results of this research will assist in developing reliable post-quantum schemes to protect information. Since Quantum Physics and security levels could be slightly difficult to understand for people without professional skills, we will need to involve specialists in Quantum Physics and cybersecurity.

II Literature Review

Niederhagen and Waidner [1] state the general concepts behind quantum computing. Along with [3], researchers suggest important tasks to enable the deployment of quantum schemes, namely the need for standardization and testing. The post-quantum encryption scheme using isogeny-based cryptography [5] will most likely resist quantum attacks. The proposed scheme achieved guaranteed IND-CCA (Indistinguishability under Adaptive Chosen Cipher-Text Attacks) and EUF-CMA (Existential-Unforgeability under Chosen-Message Attack) security and employed identity-based cryptography (IBC) for post-quantum security [5]. Although code-based hybrid encryption [8] also achieves an IND-CCA secure level, such a scheme does rely on matrix multiplication, which could make the algorithm insufficient in contrast to [5]. Therefore, QC-LDPC (Quasi-Cyclic Low-Density Parity Check) code-based encryption can potentially resist the quantum attack with lower performance than other

approaches [8].

Borges [15] performed an exploratory data analysis by assessing the classical and post-quantum security protocols as mathematical problems. The approach is similar to that used by [2]. Both works compare post-quantum cybersecurity protocols, but [15] covers a broader range of types, whereas [2] focuses only on lattice-based schemes. Mitchell [4] and Chamola *et al.* [2] considered the practical applications of quantum cryptography and discussed communication networks and schemes for maintaining a secure connection. While the research methods differ, the conclusions align. Researchers [14] solve the cybersecurity issue related to data transmission among IoT (Internet-of-Things) devices. Researchers introduce a new cybersecurity protocol QHF (Quantum Hash Function-Based Protocol) and test its' resistance to existing attacks. The algorithm [14] proved to be secure against impersonation attacks, no-message attacks, and message attacks. The next protocol is AQW (Alternate Quantum Walks), which uses optical image encryption and can resist noise and data loss attacks. Likewise, [7] addresses the same problem. Kumari *et al.* [7] considers the same attacks. However, unlike [14], [7] derived a SMHM (Signature-Based Merkle Hash Multiplication) protocol to deal with security threats, which is stable against other attacks. Borges *et al.* [15] and Mavroeidis *et al.* [16] share several features. Both showcase a comparison of cryptographic schemes. Although [16] gives a more comprehensive list of protocols and even contains a description of one named in [7]. The precedent works assessed the newly developed schemes using performance comparison and theoretical hash-key estimation. The present research contributes to the existing findings by aiming to compare algorithms' resistance practically by conducting a series of side-channel attacks to test implementation capacity.

III Research Design

Inductive research design is the most appropriate approach for the current study. This research will use quantum computers to examine the safety of existing post-quantum schemes. This research methodology will assume the simulation of side-channel attacks aimed at such schemes implemented on quantum computers. Quantum computers are available for open use by IBM [17] on the public website via the dashboard. Experimenting with existing

cryptographic algorithms to detect vulnerabilities in the most resistant scheme will take up to five months. Furthermore, two more months are required to write the paper. The authors have experience conducting research, followed by search and investigation skills. Additional experts in the field of cryptography may be involved in the experimental procedure since several selected cryptographic schemes are based on diverse approaches. We employ quantitative observations as the primary data collection process and choose schemes according to the selected articles [5], [7]–[14]. The secondary data will be the general information about elected schemes. Initially, current study consider nine post-quantum algorithms [5], [7]–[14].

The research, however, is subject to several limitations. The first is the nature of the present inquiry. Namely, the longitudinal study will not consider the algorithms proposed during the experimentation procedure. Future studies may address this limitation by extending the range of the schemes. The second limitation is the limitation to 17 articles. Nevertheless, adding new articles may avoid this limitation in future research.

IV Anticipated Results

As a result of the experimental simulation of attacks, we will obtain statistical information about tested post-quantum algorithms, such as performance, efficiency, the correlation between encrypted and original text, noise and data loss percentage, and key sensitivity. Using these findings, we will determine algorithms that are persistent to quantum attacks. We anticipate QC-LDPC code-based encryption to achieve quantum attack resistance, however, with lower performance than other approaches [8], according to received statistics. Moreover, the proposed scheme of Merkle Hash Multiplication [7] would achieve approximately the same level of quantum security as proposed by the authors. However, the absence of the experimental CPA security level of key encryption proof by Wang *et al.* [9] may be refuted in terms of simulation. Abd-El-Atty *et al.* [10] could miss various vulnerabilities for optical image encryption on a theoretical level, which the simulation procedure would find. Similarly, Kadir *et al.* [12] provided the basis for the expectation of success for such an encryption scheme in terms of experimental simulation. The stability of healthcare image encryption by Abd EL-Latif *et al.* [13] may be doubtful in contrast to [5], [7], [8]. Researchers [13] have achieved resistance to the histogram and entropy attacks only. The quantitative

analysis of the collected findings would elaborate on the list of algorithms [5], [7]–[14] that are the most resistant to side-channel attacks. The findings may contribute to the knowledge of the proposed schemes’ security level, eliminate existing vulnerabilities, and propose more secure post-quantum encryption algorithms. Due to the increase in the development of quantum computing, current encryption schemes are in danger of being compromised [6], which are used in high technology, communication, and the processing of personal data. The post-quantum encryption algorithms may resolve the low resistance of current encryption schemes to quantum attacks. Implementing the results of actual research could help develop new resistant and safe post-quantum schemes to provide safe encryption from side-channel attacks.

V Discussion

The results of actual research are particularly pertinent in light of the development of quantum computers. These results could contribute to developing resistance against hackers’ attacks and efficient post-quantum schemes to provide reliable methods for everyone in society to hide their personal information. The experiments in threat modelling for post-quantum algorithms proposed by current work can be carried out for up to seven months since some quantum computers are already available for common use. For example, these are IBM quantum computers [17], but they operate on several qubits. As a result of the experimental procedure, we will get numerical characteristics of given schemes such as performance, efficiency, and connection between initial and encrypted data [9]–[13] and decide which scheme is the best. The obtained results could be compared with [1]–[4], [6] by evaluating practical efficiency.

However, our research has several limitations. The first limitation is the appearance of new articles during our research. The second limitation is the number of articles, which is limited to 17. Extension of the range of schemes will overcome the first limitation. Expanding the number of sources will surmount the second limitation. Future research could further use more quantum computing resources to conduct a precise experimental analysis of new post-quantum algorithms. Moreover, researchers could suggest corrections for existing quantum-resistant schemes.

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